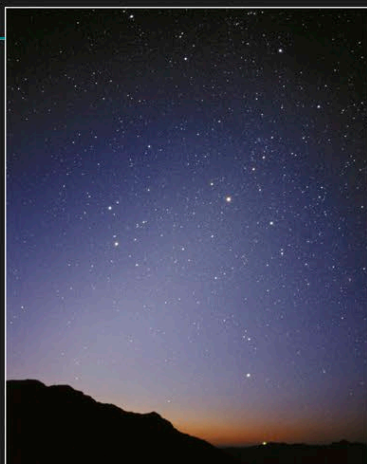




SEEING THE ZODIACAL LIGHT

The zodiacal light is most distinct just before dawn in fall, far from any light pollution. It is near the horizon and forms a rough triangle.

ZODIACAL LIGHT

A faint glow is sometimes visible in the eastern sky before dawn or occasionally in the west after sunset. Called zodiacal light, it is caused by sunlight scattered off interplanetary dust particles in the plane of the solar system—the ecliptic plane (see p. 64). The mixture of wavelengths in the light is the same as that in the Sun's spectrum. A related phenomenon is called the gegenschein (German for "counterglow"). It is sometimes perceivable on a dark night, far from any light pollution, as a spot on the celestial sphere directly opposite the Sun's position in the sky. The dust particles in space responsible for both zodiacal light and gegenschein are thought to be from asteroid collisions and comets and have diameters of about 0.04 in (1 mm).



THE GEGENSCHIN

This faint, circular glow, 10° across, is most often spotted at midnight, in an area above the southern horizon (for Northern Hemisphere viewers).

NOCTILUCENT CLOUDS

Clouds at extremely high altitude (around 50 miles/80 km up) in Earth's atmosphere can shine at night by reflecting sunlight long after the Sun has set. These "noctilucent" (night-shining) clouds are seen after sunset or before dawn. It is thought that they consist of small, ice-coated particles that reflect sunlight. Noctilucent clouds are most often seen between latitudes 50° and 65° north and south, from May to August in northern latitudes and November to February in southern latitudes. They may also form at other latitudes and times of year.

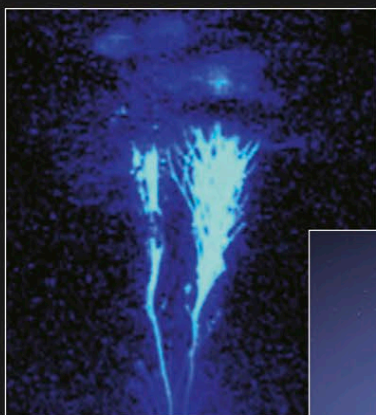
SHINING CLOUDS

Noctilucent clouds are silvery-blue and usually appear as interwoven streaks. They are only ever seen against a partly lit sky background, the clouds occupying a sunlit portion of Earth's atmosphere.



MOVING LIGHTS AND FLASHES

Many phenomena can cause moving lights and flashes across the sky. Rapid streaks of light are likely to be meteors or shooting stars—that is, dust particles entering and burning up in the atmosphere. A bigger, but very rare variant is a fireball—simply a larger meteor burning up. Slower-moving, steady, or flashing lights are more likely to be aircraft, satellites, or orbiting spacecraft. Large light flashes are usually electrical discharges connected with lightning storms. In recent years, meteorologists have named two new types of lightning: "red sprites" and "blue jets." Both are electrical discharges between the tops of thunderclouds and the ionosphere above.



BLUE JETS

These cone-shaped discharges are 30–35 miles (50–60 km) high, 6 miles (10 km) wide at the top, and result from lightning in the atmosphere ionizing nitrogen atoms, causing them to glow blue as they reemit light. In the past, blue jets may have been reported as UFOs.



PATH OF THE ISS

As the International Space Station (ISS) orbits Earth, it is visible from the ground because it reflects sunlight. This photograph of the Space Station was taken using a 60-second camera exposure, which indicates how quickly the spacecraft moves across the night sky.

MYTHS AND STORIES

UFO SIGHTINGS

Every year there are reports of unidentified flying objects (UFOs). Most of these can be accounted for by natural phenomena such as bright stars, planets, and meteors, or by man-made objects such as balloons, satellites, and aircraft. After excluding such causes, there are still unexplained cases. It would be unscientific to dismiss the possibility that these UFOs are signs of extraterrestrial visitors without further investigation—just as it would be to accept it before ruling out less exotic explanations.

FLYING SAUCER?

This object, suggestive of a flying saucer, is actually a lenticular cloud. Clouds like this are usually formed by vertical air movements around the sides or summits of mountains.

